

Hypothesis Testing

Hypothesis testing is a core technique in inferential statistics used to test assumptions about population parameters based on sample data. Depending on what is known (standard deviation, type of data, one or two groups), different statistical tests are used.

1. Single Population Mean – Known Standard Deviation (Z-Test)

Hypothesis testing is used when we want to make decisions or draw conclusions about a population based on sample data. When testing a claim about a single population mean, one of the commonly used methods is the Z-test, especially when the population standard deviation is known.

The Z-test evaluates whether the observed sample mean is significantly different from a hypothesised value of the population mean.

To perform this test, we calculate the Z-statistic, which measures how many standard errors the sample mean is away from the hypothesised population mean. The formula used is as follows:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

The metrics are as follows:

- $\bar{X}$ = Sample mean
- μ = Hypothesised population mean
- σ = Population standard deviation
- n = Sample size

The Z-statistic tells us how far (in standard error units) our observed sample mean is from the hypothesised population mean. A large absolute value of Z (either positive or negative) suggests that the observed difference is unlikely to have occurred by random chance alone, assuming the null hypothesis is true.

2. Single Population Mean – Unknown Standard Deviation (T-Test)

When the population standard deviation is not known, we estimate it using the sample standard deviation. The t-distribution is used instead of the normal distribution, especially for small sample sizes ($n < 30$). This accounts for increased uncertainty due to estimation.

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

The metrics are as follows:

- $\bar{X}$ = Sample mean
- μ = Hypothesised population mean
- s = Sample standard deviation
- n = Sample size
- Degrees of freedom (df) = $n - 1$

The Z-test is appropriate in the following conditions:

- To make inferences about the average (mean) of a single population
- The population standard deviation is known
- The population is normally distributed, or the sample size is large ($n \geq 30$)

Interpreting the Results

Once the Z-statistic is computed, we compare it with the critical value from the standard normal distribution or calculate the p-value to make a decision:

- If the p-value is less than the significance level (α), we reject the null hypothesis. This means there is sufficient evidence to suggest that the population mean is different from the hypothesised value.
- If the p-value is greater than α , we fail to reject the null hypothesis, implying that the data does not provide strong enough evidence to conclude a significant difference.

3. Population Proportion – One Sample Z-Test

The one-sample Z-test for population proportion is used when we want to test whether the observed proportion in a sample is significantly different from a hypothesised value for the population proportion.

This test helps determine whether the observed sample proportion ($\hat{p}$) is different enough from the hypothesised population proportion (p) that the difference is unlikely to be due to random sampling variability.

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

The metrics are as follows:

- $\hat{p}$ = Sample proportion
- p = Hypothesised population proportion
- n = Sample size

Use the one-sample Z-test for population proportion when:

- You are testing the proportion of a single population
- The sample size is large
- The expected number of successes in a sample should be at least five, and the expected number of failures in the sample should also be at least five

Interpreting the Results

Start by calculating the Z-value using the appropriate formula for the test. Use the standard normal distribution to find the p-value that corresponds to the calculated Z-value. Compare the p-value with the chosen level of significance:

- If the p-value is less than alpha, reject the null hypothesis. This means there is sufficient evidence to suggest that the population proportion is different from the hypothesised value.
- If the p-value is greater than or equal to alpha, do not reject the null hypothesis. This indicates that the sample data does not provide strong enough evidence to conclude that the population proportion is different from the hypothesised value.

4. Two Population Means – Known Standard Deviations (Z-Test)

This test is used to compare the means of two independent populations when the population standard deviations are known. It helps us determine whether the difference between the two-sample means is meaningful or just due to random variation in the samples.

The Z-test calculates how far the observed difference in sample means is from the hypothesised difference in population means, measured in standard error units. This tells us whether the difference is statistically significant or likely occurred by chance.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

The metrics are as follows:

- $\bar{x}_1, \bar{x}_2$ = Sample means
- μ_1, μ_2 = Hypothesised means
- σ_1, σ_2 = Known population standard deviations
- n_1, n_2 = Sample sizes

The Z-test for comparing two population means is valid under the following assumptions:

- The two samples are randomly selected and independent of each other
- The population standard deviations are known
- Each population is normally distributed, or the sample sizes are large ($(n_1 \geq 30)$ and $(n_2 \geq 30)$)

Interpreting the Results

First, compute the Z-value using the formula. Then, find the corresponding p-value from the standard normal distribution. Compare the p-value to the significance level α :

- If the p-value is less than α , reject the null hypothesis. This means the observed difference in sample means is statistically significant and unlikely to have occurred by chance.
- If the p-value is greater than or equal to α , do not reject the null hypothesis. In this case, the data does not provide sufficient evidence to conclude that the population means are different.

5. Two Population Means – Unknown Standard Deviations (Z or T-Test)

This test is used to compare the means of two independent populations when the population standard deviations are not known. When both sample sizes are large (typically 30 or more), the Z-test can still be applied using sample standard deviations in place of population values. However, when the sample sizes are small, a two-sample t-test is used.

This test helps us assess whether the observed difference between the two-sample means is large enough to conclude that a real difference exists between the population means.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The metrics are as follows:

- $\bar{x}_1, \bar{x}_2$ = Sample means
- μ_1, μ_2 = Hypothesised population means
- s_1, s_2 = Sample standard deviations
- n_1, n_2 = Sample sizes

The Z-test for comparing two population means is valid under the following assumptions:

- When you are comparing the means of two independent populations
- The population standard deviations are unknown
- If the sample sizes are large (typically $n_1 \geq 30$ and $n_2 \geq 30$), a Z-test can be used
- If the sample sizes are small, and the data is approximately normally distributed, use a two-sample t-test.

Interpreting the Results

First, calculate the test statistic using the formula. Then, determine the p-value using either the standard normal distribution in the case of a Z-test or the t-distribution if it is a t-test. compare the p-value with the significance level (α):

- If the p-value is less than (α), reject the null hypothesis. This means there is a statistically significant difference between the two population means.
- If the p-value is greater than or equal to (α), do not reject the null hypothesis. This means the data does not provide enough evidence to conclude that the population means are different.

6. Two Population Means – Matched Pairs (Paired t-Test)

The paired t-test is used when you are comparing two related or dependent samples. This typically occurs in before-and-after scenarios, measurements taken on the same subject under different conditions or matched subjects in controlled experiments. The objective of this test is to determine whether the mean of the differences between paired observations is significantly different from zero.

Instead of analysing two samples separately, the paired t-test focusses on the differences between each pair and treats these differences as a single sample. The null hypothesis assumes that the mean of these differences is zero, indicating no change or effect.

The mean of differences is expressed by the following equation:

$$\bar{d} = \frac{\sum_{i=1}^n d_i}{n}$$

The standard deviation of differences is expressed by the following equation:

$$s_d = \sqrt{\frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n - 1}}$$

t-Statistic:

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

The metrics are as follows:

- d_i = Paired difference
- $\bar{d}$ = Mean of differences
- s_d = Standard deviation of differences
- n = Number of pairs
- $\mu_d = 0$ under null hypothesis
- $df = n - 1$

The paired t-test is appropriate under the following conditions:

- You are working with two related samples or repeated measurements on the same units.
- The differences between the paired values are approximately normally distributed.
- The population standard deviation of the differences is unknown.
- The sample of paired differences is random, and each pair is independent of the others.

Interpreting the Results

First, calculate the mean and standard deviation of the differences using the given formulae. Next, compute the t-statistic and find the p-value using the t-distribution. Compare the p-value to the significance level (α):

- If the p-value is less than α , reject the null hypothesis. This means there is statistically significant evidence that the mean difference is different from zero.

- If the p-value is greater than or equal to α , do not reject the null hypothesis. This means there is not enough evidence to suggest a significant difference between the paired observations

7. Two Proportions – Z-Test for Proportion Difference

The Z-test for two proportions is used to compare the proportions of a certain outcome between two independent groups. It helps determine whether the observed difference between the sample proportions is statistically significant or if it could have occurred by random chance.

$$z = \frac{\widehat{p}_1 - \widehat{p}_2}{\sqrt{\widehat{p}(1 - \widehat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \qquad \widehat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

The metrics are as follows:

- $\widehat{p}_1, \widehat{p}_2$ = Sample proportions
- $\widehat{p}$ = Combined proportion
- n_1, n_2 = Sample sizes

The Z-test for two proportions is valid under the following conditions:

- The samples are independent of each other.
- Each sample is random and sufficiently large for normal approximation.
- The following conditions must be satisfied for both groups:

$$n_1\widehat{p}_1 \geq 5, \quad n_1(1 - \widehat{p}_1) \geq 5$$

$$n_2\widehat{p}_2 \geq 5, \quad n_2(1 - \widehat{p}_2) \geq 5$$

Interpreting the Results

Begin by calculating the test statistic using the formula. Then, determine the p-value using the standard normal distribution.

Compare the p-value to your chosen significance level:

- If the p-value is less than α , reject the null hypothesis. This means there is statistically significant evidence that the two population proportions are different.
- If the p-value is greater than or equal to α , do not reject the null hypothesis. This means the data does not provide strong evidence that the population proportions differ.

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